Is the Platform Part of the Measurement?
A Protocol and Simulation Study for Cross-Platform Equivalence of Machine-Labelled Policy Sentiment

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Abstract

Comparing machine-labelled sentiment across social media platforms is now routine: differences in classifier scores between platforms are read as differences in public opinion. These comparisons rest on an assumption that the classifier measures the same construct, on the same scale, on every platform. To our knowledge, no study has tested that assumption for machine-labelled constructs by asking whether the platform is part of the measurement. We adapt the psychometric equivalence toolkit (generalizability theory, measurement invariance and differential item functioning, and score linking) to machine-labelled policy sentiment, treating the platform as a survey mode, and specify a six-step protocol with anchor events fixed in advance, an anchor-density audit, and a two-classifier convergence control. A simulation with known ground truth quantifies what the assumption costs in one generating configuration fixed in advance: across 500 replications, a pooled index that takes platform scores at face value misstates a known event shift by roughly an eighth, and because the platforms measure through different functions it reports the wrong sign of aggregate sentiment on nearly a quarter of days. A measurement-adjusted index recovers the shift with negligible bias and the correct sign on all but one day in sixty on average. A case study of the instrument from the first author's MSc dissertation, a fine-tuned BERT classifier over 255,446 UK economic-policy comments from Twitter/X, Reddit, YouTube, and Quora, motivates the design and illustrates the instrument-dependence the protocol addresses. The full four-platform application, under fresh collection with the pre-specified design, is the designated follow-up study. The protocol lets a researcher test whether pooling platforms is defensible in a given corpus, or whether platform scores must be linked before any average is taken.

1 Introduction

Computational social science now compares machine-labelled text across platforms as a matter of course. Overgaard et al. [2024] apply a BERT classifier to Facebook and Twitter posts and read the difference in scores as a difference in behaviour. Wei et al. [2026] run the same design across Douyin and TikTok. Studies in this mould publish a cross-platform comparison, or a pooled multi-platform index, on the assumption that the classifier behaves identically wherever it is pointed. The assumption is rarely stated, still more rarely examined, and it is not an obviously safe one. Nogara et al. [2025] report that Perspective API scores German comments as more toxic than equivalent English ones: the instrument behaves differently in different languages. In Atreja et al. [2025], LLM

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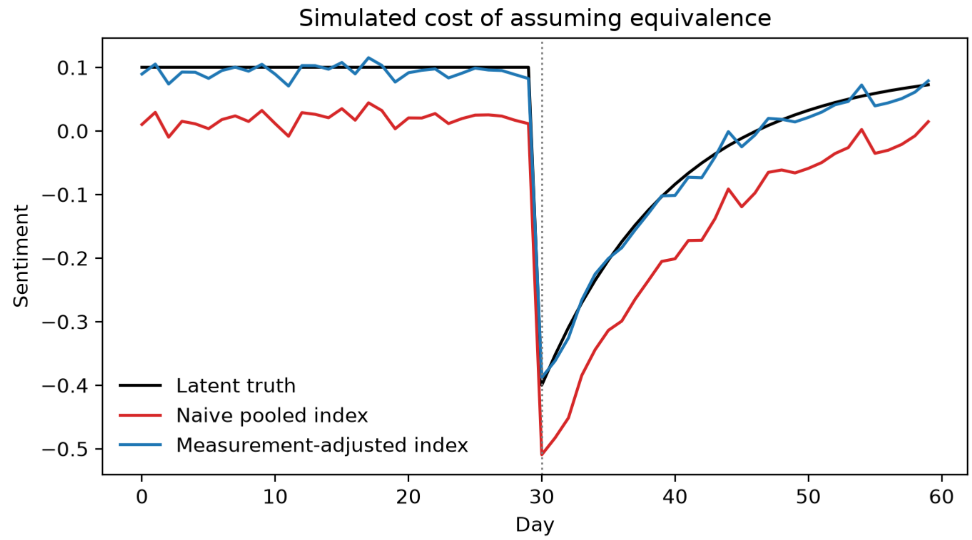

Figure 1: What assuming equivalence costs, in one simulation run with known ground truth (seed 20261201; parameters fixed ex ante, shipped with the code). The naive pooled index misstates the event shift and reports the wrong sign of aggregate sentiment on 13 of 60 days in this run; the measurement-adjusted index tracks the latent truth with no wrong sign day. Here the true event shift is -0.332; the naive index estimates -0.369 and the adjusted index -0.326. Section 7 reports 500 replications.

annotations shift under changes of prompt wording alone. And Greene et al. [2025] find that the same political actors are read differently on different platforms. Classifiers are instruments, and their behaviour depends on the conditions in which they are used.

When the labels are machine-produced, nobody asks with a formal test whether the platform is part of the measurement. When a classifier scores Reddit discussion of a budget more negative than tweets about the same budget, the difference is read as a fact about opinion, when it could equally be a property of the instrument. Character limits, anonymity, threading, moderation, and audience shape how the same attitude is expressed, and therefore how it is scored. Sen et al. [2021] give this hazard a name, platform-affordance error, in their total error framework for digital traces, but they provide no formal test for it. Recent correction methods repair label error against a human gold standard [Egami et al., 2023] and probe the fragility of LLM annotation pipelines [Baumann et al., 2025, Halterman and Keith, 2026], yet none of them models the platform as a source of measurement error.

Survey methodology met a structurally identical problem decades ago and built machinery for it. The same respondent answers differently by telephone, on paper, and face to face; the field calls this a mode effect and does not treat cross-mode differences as substantive until measurement equivalence has been established [Tourangeau et al., 2000, Klausch et al., 2013, Hox et al., 2015], within a total-error accounting of where distortions enter [Groves and Lyberg, 2010]. Our claim is that a platform is a mode. The analogy has limits: absent random assignment of users to platforms or within-user cross-posting, the design identifies platform-level non-equivalence, a bundle of selection and measurement that Section 8 unpacks, with mode as the leading interpretation rather than an established fact. Affordances stand to posts as interview settings stand to answers, and the psychometric toolkit built for modes and groups (generalizability theory [Cronbach et al., 1972, Brennan, 2001], measurement invariance testing [Meredith, 1993, Putnick and Bornstein, 2016], and score linking [Kolen and Brennan, 2014]) transfers to machine-labelled constructs with the platforms as the groups. The toolkit has reached survey self-reports [Boyd et al., 2024], annotator facets [Kennedy et al., 2020, Sachdeva et al., 2022], and single-platform text scales [Pokropek, 2026], but the platform itself has stayed an unmodelled stratum.

The question is also practical. Since the closure of the major research APIs, few teams can study one platform at depth; instead they assemble patchworks of whatever platforms remain accessible, with heterogeneous coverage and access terms [Davidson et al., 2023, Goanta et al., 2026]. Every such patchwork implicitly pools or compares scores across platforms. If a unit of classifier score does not mean the same thing on Reddit as on X, the pooled trend lines and cross-platform contrasts now filling the literature carry an error term of unknown size.

This paper makes four contributions.

1. We formalise the platform-as-mode question for machine-labelled constructs and give what we believe is the first adaptation of the psychometric equivalence toolkit to machine-classified policy sentiment with platform as the grouping facet.

2. We specify a six-step protocol (Sections 4 to 7): same-event divergence testing, a fixed-facet generalizability study against a human-validated criterion, categorical invariance and DIF under effect-size criteria, multilevel adjustment for composition and timing, anchor-based linking centred on the September 2022 mini-budget, and a simulation of the cost of skipping the first five steps. Every test is replicated with a non-lexicon transformer, so only convergent non-invariance is read as a mode effect [Czarnek and Stillwell, 2022], and invariance results are treated as diagnostic evidence rather than grounds to pool [Robitzsch and Lüdtke, 2023].

3. We quantify the cost of assuming equivalence in a simulation with known ground truth (Section 7, Figure 1): across 500 replications of one generating configuration fixed in advance, a naive pooled index misstates a known event shift by roughly an eighth and, because the platforms measure through different functions, reports the wrong sign of aggregate sentiment on nearly a quarter of days, errors the measurement-adjusted index avoids.

4. We ground the protocol in the instrument from the first author's MSc dissertation (Section 3), a fine-tuned BERT classifier over 255,446 UK economic-policy comments from Twitter/X, Reddit, YouTube, and Quora, whose execution records supply the motivating evidence; the full four-platform application, under fresh collection with the pre-specified design, is the designated follow-up study.

2 Related Work

Mode effects in survey measurement. Survey methodology settled long ago that the instrument is part of the measurement. Answers to the same question differ by mode of administration because each mode changes the cognitive and social conditions of answering [Tourangeau et al., 2000]. Klausch et al. [2013] showed that telephone, face-to-face, and web modes yield non-equivalent measurement of identical attitude scales, and mixed-mode designs now model mode explicitly instead of ignoring it as a nuisance [Hox et al., 2015]. The Total Survey Error framework gives these effects a named place in the error budget [Groves and Lyberg, 2010]. Sen et al. [2021] transported this thinking to digital traces and explicitly identified platform-affordance error as the trace analogue of mode error, but the framework names the error without supplying a test for it. Our premise is simply that platform should receive the same formal treatment for machine-labelled trace data that mode receives for surveys.

Measurement for machine-labelled text. A second literature treats classifiers as measurement instruments. Jacobs and Wallach [2021] argue that computational pipelines operationalise unobservable constructs through measurement models whose validity must be tested, and Halterman and Keith [2026] extend this framing to large language models. Empirical work in this vein audits the instrument itself. Pellert et al. [2022] validate sentiment macroscopes against self-report. Atreja et al. [2025] show that model-generated labels shift with prompt design in unpredictable ways, and Baumann et al. [2025] show that defensible configuration choices alone can flip downstream conclusions. Egami et al. [2023] correct downstream estimates for classifier error with design-based supervised learning, but the correction targets label error against a gold standard and leaves platform-specific distortion untouched. In all of this work the classifier is scrutinised; the platform it reads from is not.

Cross-platform studies and assumed equivalence. Multi-platform designs have become standard, and the post-API turn has made heterogeneous data regimes the norm [Davidson et al., 2023, Goanta et al., 2026]. Overgaard et al. [2024] apply a single BERT classifier across Facebook and Twitter and compare the resulting scores directly; Wei et al. [2026] compare Douyin and TikTok in the same way. There is growing evidence that this is risky. Greene et al. [2025] find that the same actors are read differently across platforms, and Nogara et al. [2025] show that Perspective API scores German-language text as more toxic, a measurement artefact that masquerades as a substantive difference. The field's classic cautions concern populations and samples: who is on the platform [Tufekci, 2014], how the stream is sampled [Morstatter et al., 2013], whether models generalise beyond their training population [Cohen and Ruths, 2013], and whether text sentiment tracks opinion at all [O'Connor et al., 2010]. The complementary question, whether the same score means the same thing on Reddit as on YouTube, is routinely assumed and almost never tested.

The psychometric toolkit and nearest misses. The tools for this question already exist in psychometrics. Generalizability theory partitions score variance among measurement facets [Cronbach et al., 1972, Brennan, 2001]; measurement invariance testing asks whether indicators relate to a construct identically across groups [Meredith, 1993, Putnick and Bornstein, 2016]; linking places scores from different instruments on a common scale [Kolen and Brennan, 2014]; and invariance results are read as diagnostic evidence rather than permission to compare [Robitzsch and Lüdtke, 2023]. Several recent studies come close to combining this toolkit with platform data without quite doing so. Boyd et al. [2024] do test cross-platform invariance, though of survey self-reports rather than machine labels. Faceted Rasch measurement has reached annotation through Kennedy et al. [2020] and Sachdeva et al. [2022], with annotators as the modelled facet and platform left out of the model. In Pellert et al. [2022] validation is by correlation alone, with no invariance structure. The multitrait-multimethod models of Cernat et al. [2025] compare survey with digital-trace measures rather than platform with platform. Han and Anderson [2026] document platform-dependent selection and measurement bias, but in online review ratings rather than machine-labelled text. Pokropek [2026] fit factor models to text indicators while grouping by time within a single platform. And generalizability designs have reached LLM annotation, with prompt and model choice as variance facets, without yet reaching platform [Camuffo et al., 2026].

To our knowledge, no study has applied the psychometric equivalence toolkit of generalizability, invariance, and linking to machine-classified policy sentiment with platform as the grouping facet, and this paper specifies that test.

3 The Motivating Case and the Pre-Specified Application

This section describes the corpus and instrument from the first author's MSc dissertation (University of Portsmouth, 2025) that motivate the paper, and the validation design the pre-specified application will execute. The account rests on the dissertation's execution records, its code, pipeline logs, and fully executed notebook; Section 8 documents the discrepancies. Dissertation-stage figures are reproduced in their original form throughout, as evidence of the instrument as it stood; Appendix B states the figure standard the application will apply to its fresh counterparts.

Corpus. The dissertation-stage corpus comprised 255,446 public comments attached to 6,650 parent posts, about UK economic policy (per the pipeline records; the dissertation's own headline of approximately 279,000, p. 25, matches none of them). It was collected from Twitter/X (licensed third-party archive, January 2018 to July 2025; dissertation p. 21), Reddit (official developer API, no stated date restriction; p. 22), YouTube (Data API v3, top-level comments on major UK news and government channels; pp. 22-23), and Quora (the same licensed analytics partner as Twitter; p. 23). The precise collection window cannot be reconstructed: the dissertation scopes the study to 2010 to mid-2025 (p. 7), gives January 2018 to July 2025 for Twitter only (p. 21), states no restriction for Reddit (p. 22), and its forecast figures span 2010 to 2026 (pp. 38-39); the application fixes its window ex ante instead.

Collection was organised by an auditable keyword scheme with six seed-rule topics: taxation and fiscal policy; inflation, cost of living and prices; welfare and benefits; energy and utilities; housing and mortgages; and the labour market (dissertation p. 18; the scheme, including its disambiguation priority and negation window, is transcribed in Appendix C). Keyword tagging then assigned classified posts to nine analysis categories, of which an Other/Unclear category absorbed roughly a third (dissertation pp. 41-42). Per-category counts are recorded in the executed figure (the N labels of Figure 2). The unit of analysis is the comment, with parent posts available as a clustering level. The execution records do not separate the corpus by platform: the dissertation's per-platform figures (approximately 120k Twitter, 130k Reddit, 120k YouTube, 25k Quora; pp. 21-24) sum to roughly 395k, matching neither the 279k headline nor the 255,446 pipeline total. The application reports its composition from collection logs by construction.

Machine labels came from a fine-tuned BERT classifier (macro-F1 0.878 against the VADER-seeded corpus labels, not against the human criterion; dissertation Table 7.1, p. 35), trained on VADER-seeded labels; Table 1 reproduces the dissertation-stage comparison and Figure 2 the per-category sentiment shares it produced. We disclose this circularity and address it with the human criterion below and with the artefact control in Section 5. The dissertation-stage shares themselves fail face validity: every policy category is plurality positive for UK economic discourse spanning a cost-of-living crisis, which is substantively implausible and is best read as a candidate artefact of the lexicon-seeded labels. The expanded human criterion measures this directly; if it confirms a positive bias, that bias enters the generalizability and invariance analyses as a measured quantity rather than an assumption. The application holds its corpus fixed across every corpus-based step, so every comparison uses the same comments.

Anchor events. Cross-platform comparison requires posts about the same event at the same time. We fixed fifteen UK fiscal policy events (Table 2) on 21 August 2026, before any window count was computed, with the September 2022 mini-budget as the centrepiece, and defined two windows

Table 1: Dissertation-stage classifier comparison (dissertation Table 7.1, p. 35). The BERT and BiLSTM rows match the appendix confusion matrices exactly and therefore measure agreement with the VADER-seeded labels over the full classified corpus; the VADER and logistic regression rows appear to be scored against the 200-post human test set. The rows do not share a reference standard, and no row measures fine-tuned performance against the human criterion; the dissertation's own notebook output computes the BERT figures against the corpus labels (p. 31). Values are shown as printed in the dissertation. The application will retrain both families by the surviving procedure on its own corpus and score every model against both reference standards, with binomial intervals on the human-criterion figures.

Model	Accuracy	Macro precision	Macro recall	Macro F1
BERT (fine-tuned)	88.06%	0.8815	0.8744	0.8778
BiLSTM (GloVe)	81.51%	0.8141	0.8099	0.8111
VADER baseline	65.20%	0.66	0.65	0.65
Logistic regression (BoW)	71.40%	0.72	0.71	0.71

per event: an acute 48-hour window and an extended two-week window. The application's event-assignment rule is fixed ex ante: the keyword scheme of Appendix C applied to the comment text and parent title, with the platform's comment timestamp as the time field, and relevance validated on a manual subsample within the expanded annotation. We expect the acute windows to be sparse on Quora, whose question-and-answer format does not lend itself to event-locked measurement; the audit is designed to identify such platforms so they can be given a reduced role or excluded, and when a platform is excluded we report a gap in the data and claim nothing about its equivalence. An anchor-density audit precedes all analysis (Section 5); platforms that fail the minimum post-count threshold for an event-window are excluded from that comparison, and the analysis is restricted to the platforms with adequate density if necessary. The dissertation reports lower data density in early 2022 (p. 25), so density is checked per event-window rather than assumed.

Gold-standard sample. The dissertation validated machine labels on 200 manually annotated posts, balanced across the four platforms, with inter-annotator agreement of Cohen's $\kappa \approx 0.78$ (dissertation pp. 25, 28). Because that sample is small and was not drawn by probability sampling, the pre-specified application expands it to 1,000-1,500 comments drawn by stratified probability sampling, with strata defined by platform, anchor event, and predicted class, and inclusion probabilities retained for design-weighted estimates. A second annotator codes an overlap subset, targeted at a quarter to a third of the sample; disagreements are adjudicated and inter-annotator reliability is reported from the pre-adjudication ratings. This sample is the criterion for the generalizability study and for validating the machine labels; the invariance models themselves are fitted on the machine-labelled corpus within anchor windows, with the criterion validating their inputs rather than forming their estimation sample. Because the criterion rests on human judgement rather than on a second lexicon, it breaks the circularity of the VADER-seeded training labels. Human validation remains essential for machine annotation [Pangakis and Wolken, 2025, Atreja et al., 2025].

Ethics. The underlying study was reviewed by the University of Portsmouth Faculty Ethics Committee (School of Computing) and classified as low risk (certificate TETHIC-2025-111094; dissertation pp. 15, 60), with legitimate interests as the lawful basis under UK GDPR (p. 14). We

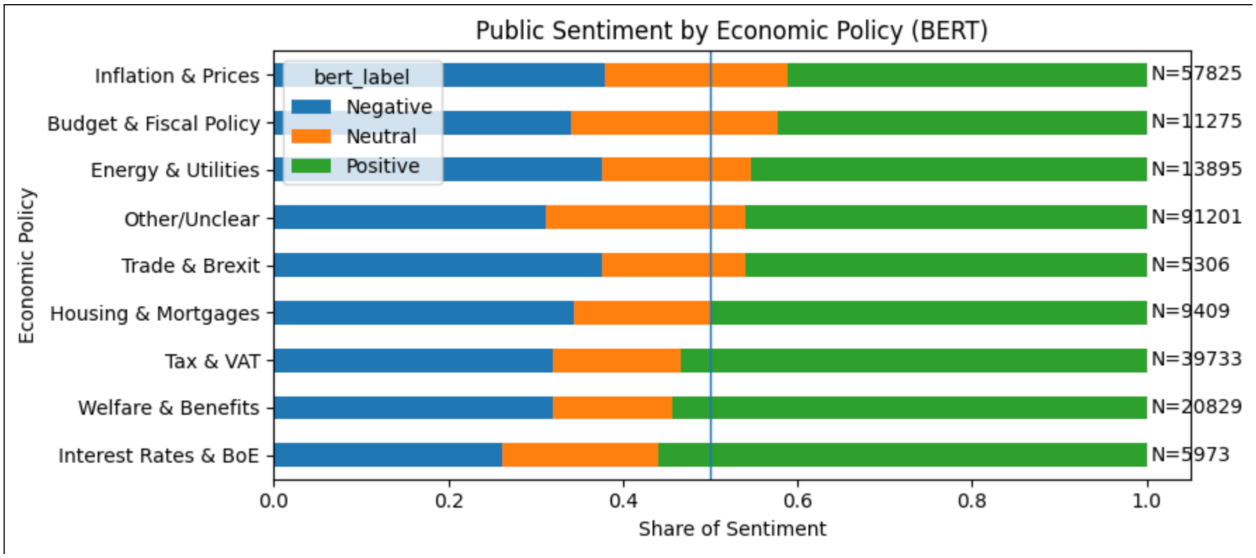

Figure 2: Sentiment shares by policy category under the fine-tuned BERT classifier, as they stood at dissertation stage (Figure 7.12, p. 41). The nine keyword-assigned categories carry per-category counts that sum to 255,446, the total the surviving pipeline records establish, and the Other/Unclear category absorbs roughly a third of classified posts.

analyse public posts only, apply data minimisation, and process data under the Article 89 research safeguards. We report aggregates, paraphrase any quoted material, make no attempt to identify users, and, in the application, share dehydrated identifiers rather than full text where platform terms permit. The dissertation-stage corpus is not redistributed; its identifiers were hashed at collection (p. 25).

4 Step 1: Establishing Divergence

Before asking whether the platform shapes the measurement, we establish that there is something to explain. For each anchor event and window, we compute the distribution of machine-assigned sentiment labels per platform and compare platforms pairwise. Because the comparison is confined to a shared event window, it constrains topic and time far more tightly than whole-corpus comparison, though different sub-events may still dominate on different platforms within a window; within-window composition is handled in Step 4. We test differences in class proportions with chi-square tests and report bootstrap confidence intervals, clustered by user or thread where hashed identifiers permit, and Cramér's V, since at this sample size significance alone is uninformative and interpretation rests on the effect sizes. Divergence at this step is only a precondition. It is consistent both with genuinely different populations and with platform-dependent measurement, and the remainder of the paper separates the two. Prior work stops at this step or assumes it away [Pellert et al., 2022, Overgaard et al., 2024].

This step executes in the pre-specified application, on its own corpus; the simulation of Section 7 demonstrates, with known ground truth, what is at stake when the equivalence question is not asked at all.

Table 2: Anchor events fixed on 21 August 2026, before any window count was computed. Per-platform counts in the 48-hour and two-week windows are produced by the application's density audit on its corpus; the inclusion thresholds, fixed at the same time, are 200 comments per platform-event cell in the two-week window and 50 in the 48-hour window. The audit re-verifies each date against official records before counting.

Event	Date
Budget 2020 (Covid response)	11 Mar 2020
Furlough scheme announced	20 Mar 2020
Budget 2021	3 Mar 2021
Autumn Budget 2021	27 Oct 2021
Energy bills package and Bank rate rise	3 Feb 2022
Cost-of-living package and windfall tax	26 May 2022
Energy Price Guarantee	8 Sep 2022
Mini-budget (centrepiece anchor)	23 Sep 2022
Statement reversing the mini-budget	17 Oct 2022
Autumn Statement 2022	17 Nov 2022
Spring Budget 2023	15 Mar 2023
Autumn Statement 2023	22 Nov 2023
Spring Budget 2024	6 Mar 2024
Autumn Budget 2024	30 Oct 2024
Spring Statement 2025	26 Mar 2025

5 Steps 2 to 4: Reliability, Invariance, Adjustment

We treat the classifier as a measurement instrument and the platform as a candidate component of the measurement procedure [Jacobs and Wallach, 2021, Halterman and Keith, 2026]. The design proceeds in six steps: establish divergence (Section 4); estimate reliability per platform with a generalizability study; test measurement invariance and differential item functioning with platform as the grouping facet; adjust for composition and timing in a multilevel model; link scales where the evidence permits (Section 6); and quantify the cost of skipping the first five steps (Section 7).

Two design commitments apply to every step that follows. The anchor-density audit comes first: every event-window-platform cell must meet a minimum post count before it enters any model, and if cells fail, the study narrows to the dense subset of platforms rather than papering over sparsity [Morstatter et al., 2013]. And platform is a fixed facet throughout: our claims concern these four platforms, not a universe of platforms from which they were sampled.

5.1 Step 2: Fixed-facet generalizability study

The criterion is the gold-standard sample of Section 3. For each platform m we estimate a generalizability study with posts nested in events and coders as a random facet, analysed on pre-adjudication ratings; the design is unbalanced by construction, since the second coder rates a subset, and components are estimated by REML with interval estimates rather than point comparisons. The classifier enters the same decomposition as an additional fixed rater, so machine-human disagreement is an estimable per-platform component:

$$X^{(m)}_{p(e)c} = \mu^{(m)} + \nu^{(m)}_{e} + \nu^{(m)}_{p(e)} + \nu^{(m)}_{c} + \nu^{(m)}_{ec} + \nu^{(m)}_{p(e)c,\varepsilon}, \tag{1}$$

with variance components for events, posts within events, coders, their interactions, and residual estimated separately at each fixed level m [Cronbach et al., 1972, Brennan, 2001]. Comparing absolute error components across platforms shows whether the measurement procedure is equally dependable on every platform, something a single reliability coefficient cannot show. Dependability coefficients are population-bound, so cross-platform coefficient comparisons are reported as descriptive only. Generalizability designs have recently modelled prompts and models as facets of automated annotation [Camuffo et al., 2026], and annotator-facet studies exist [Kennedy et al., 2020, Sachdeva et al., 2022], but none of them models the platform itself.

5.2 Step 3: Invariance and DIF with platform as the group

We fit multiple-group measurement models grouped by platform. The indicators are binary presence codes for each of the eight NRC emotions (Figure 3), derived from the raw lexicon match counts rather than a single dominant label (the executed notebook demonstrates this computation per comment; Figure 11), together with the three-category sentiment class treated as ordinal under the threshold parameterisation. Dimensionality is tested per platform before any invariance constraint is imposed, and the invariance battery is conditional on an acceptable configural model. Because the NRC indicators derive from a lexicon, they are excluded from the non-lexicon replication arm, which is restricted to the sentiment indicator set.

Four safeguards apply at this step. Identification follows the threshold parameterisation for categorical indicators [Wu and Estabrook, 2016]; we do not treat categorical labels as linear indicators. Invariance decisions rest on effect-size criteria, changes in CFI and RMSEA [Chen, 2007, Putnick and Bornstein, 2016]; chi-square values are reported but decisions do not rest on them, since at this sample size the test rejects trivial differences. Because these cutoffs were calibrated for continuous indicators under maximum likelihood, we treat them as heuristics subject to sensitivity analysis under the categorical estimator, and per-event DIF results aggregate into a platform-level claim by a rule fixed before estimation: an indicator-platform pair is flagged only when non-invariance appears in at least two events that pass the density audit, in both classifier arms; for the NRC emotion indicators, which exist only in the lexicon arm, the two-event rule applies within that arm and any flag is reported as lexicon-conditional. DIF is tested separately within each anchor event, so an event-specific artefact cannot be mistaken for a global platform effect. And because the choice of anchors can itself build in the conclusion, we re-estimate DIF anchor-free [Chen et al., 2023] and run alignment optimisation [Muthén and Asparouhov, 2018] alongside the constrained models. These are three identification strategies for the same model rather than independent tests, so agreement rules out anchor-choice artefacts and nothing more; a claim of non-invariance must survive all three, a claim of invariance must do the same, and disagreement is reported as inconclusive.

5.3 Step 4: Multilevel adjustment

Steps 2 and 3 could still confound measurement with composition. We therefore fit a multilevel model with posts nested in events and platform as a fixed effect. Time since event enters the model explicitly: platforms differ systematically in when their users respond, and this timing component may reflect real opinion dynamics as well as affordance. We therefore estimate platform contrasts with and without the latency term and report the pair, as total and direct platform differences, alongside rather than inside the measurement comparison. Where the multilevel structure permits, we also apply the cluster-bias test of Jak et al. [2013], which examines invariance violations across clusters directly.

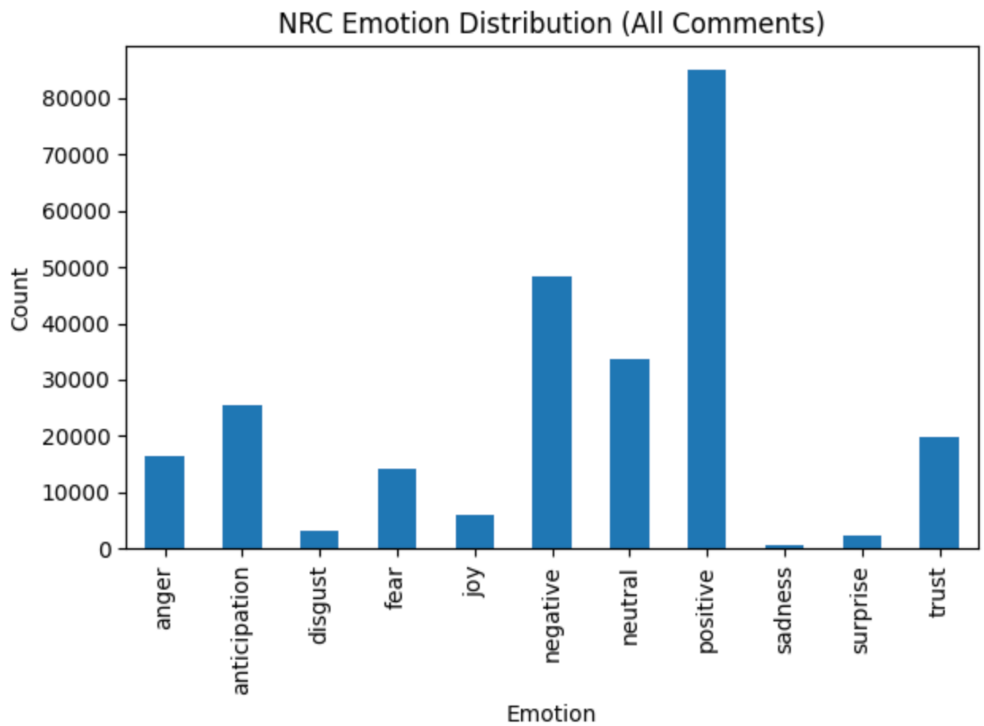

Figure 3: Dissertation-stage NRC distribution over all comments (dissertation Figure 7.8, p. 40): eight emotion categories plus positive, negative, and neutral tallies. The bars sum to the classified corpus, so each comment carried one mutually exclusive label at dissertation stage; the application replaces this single-label assignment with binary presence codes per emotion, which become the Step 3 indicators.

5.4 Classifier-artefact control and interpretive stance

A platform effect estimated with one classifier could be an artefact of that classifier, and our training labels descend from a lexicon, as already disclosed. Every sentiment-indicator analysis is therefore replicated with a non-lexicon transformer, an off-the-shelf model fine-tuned on independently human-labelled sentiment data with no VADER ancestry (a Twitter RoBERTa of the kind released by CardiffNLP fits the requirement), and only convergent non-invariance, the same pattern under both classifier families, is interpreted as platform-level non-equivalence rather than a lexicon artefact [Czarnek and Stillwell, 2022]. Convergence cannot rule out biases shared by both instrument families, a residual limitation stated in Section 8. Divergent patterns are reported as classifier artefacts, which also constrains the analyst degrees of freedom that make automated-annotation pipelines fragile [Baumann et al., 2025, Atreja et al., 2025]. The two families do behave differently: at dissertation stage the two families agreed on 88.1 per cent of comments (225,010 of 255,446; executed notebook output, and Figure 10), a figure the dissertation text misreports as 85 per cent, with disagreement varying modestly by topic, from roughly 10 to 13 per cent and highest for Trade and Brexit (Figures 4 and 5). The convergence criterion relies on this variation. The consequence for measurement is visible in Figure 6: rescoring the same corpus with the BiLSTM preserves the ranking of policy categories but changes the measured sentiment shares, a within-platform instance of the instrument-dependence this paper tests between platforms.

Finally, invariance evidence is diagnostic, not a licence [Robitzsch and Lüdtke, 2023]. Passing a

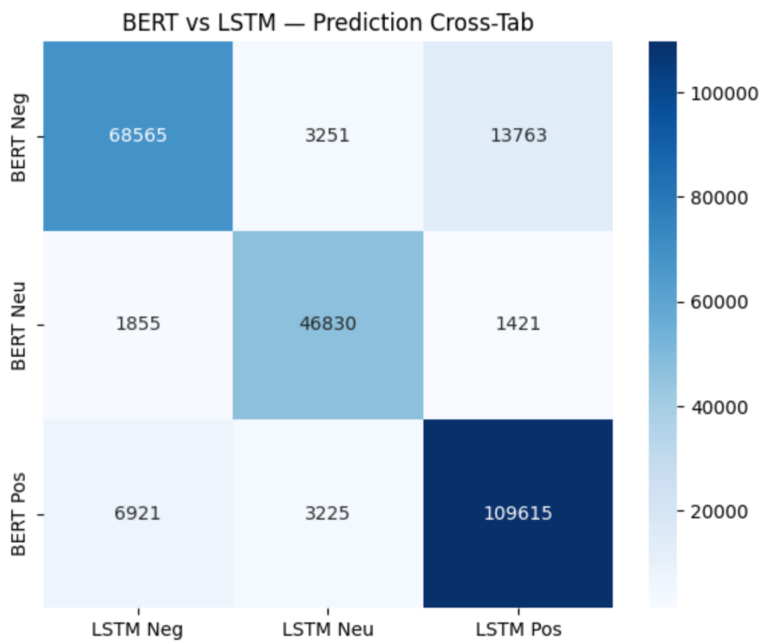

Figure 4: Dissertation-stage cross-tabulation of BERT and BiLSTM predictions over the classified corpus (dissertation Figure 7.15, p. 43). Both families were trained on the same VADER-seeded labels, so their disagreement illustrates instrument-dependence within a shared label heritage; the BiLSTM is not the non-lexicon control, which is specified in the text.

threshold does not certify pooled scores as interchangeable; it localises where comparability holds and where it fails, and every linking claim in Section 6 is conditioned on that localisation. The framework thereby formalises the platform-affordance error that Sen et al. [2021] describe but do not test, and reaches a failure mode that label-error corrections do not touch, because those correct classifier error against a criterion within a platform, not non-equivalence between platforms [Egami et al., 2023].

6 Step 5: Linking

If partial invariance holds, we link the platform scales. What we attempt is linking rather than equating: equating requires that transformed scores be interchangeable, a standard machine-labelled sentiment across platforms cannot meet, so we settle for the weaker relation [Kolen and Brennan, 2014]. The linked object is a platform-level quantity, the platform-specific latent means and variances under the partial-invariance model, not any per-post score. The pre-identified fiscal events act as common stimuli rather than common items: different posts by different authors are scored on each platform, which in Kolen and Brennan's taxonomy is a moderation design, the weakest linking category, and we label it as such. The link is identified only under the assumption that latent sentiment towards each anchor event is equal in expectation across platform populations after the Step 4 composition adjustment; where that assumption is doubted, the crosswalk removes population differences along with measurement differences, and we report both indices for that reason. The September 2022 mini-budget is the principal anchor, with the 48-hour and two-week windows defining the linking samples. Purification is ex ante: the rule for screening anchors for differential functioning is fixed before any linking function is estimated: the link is estimated, the

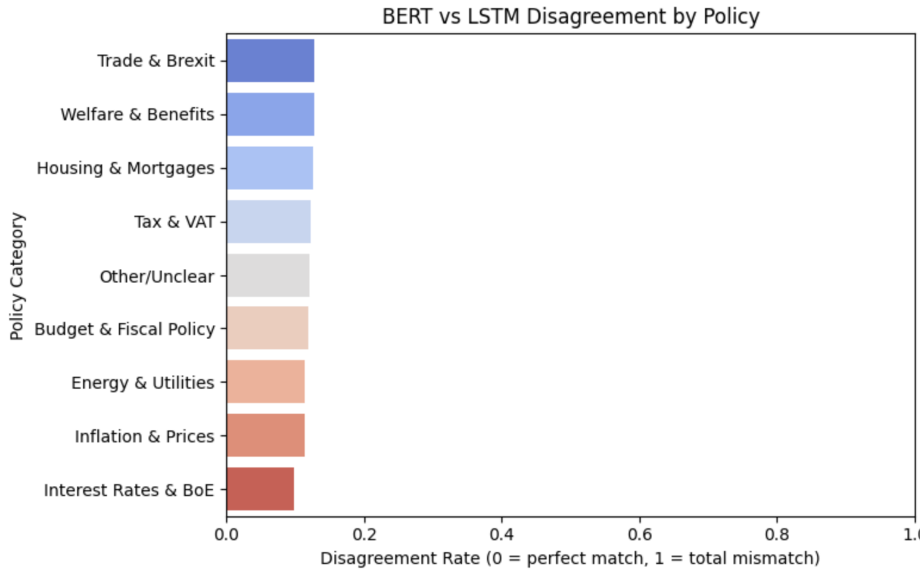

Figure 5: BERT-versus-BiLSTM disagreement rate by policy category at dissertation stage (Figure 7.14, p. 43). Disagreement is highest for Trade and Brexit and Welfare and Benefits, though the spread across categories is modest, roughly 10 to 13 per cent.

anchor event with the largest standardised DIF statistic is removed, and the link is re-estimated until no statistic exceeds its threshold, with every removed event reported. Anchors therefore cannot be chosen in a way that favours a particular result. For sensitivity we re-estimate the linking function dropping each anchor event in turn and report the spread; a link that depends on a single event is reported as such; we do not smooth over it.

The protocol does not presuppose that linking will succeed. If Step 3 shows non-invariance too severe to support even partial-invariance linking, we do not link. The deliverable then becomes platform-specific sentiment series plus the demonstration that a pooled cross-platform index is unsupported for this construct. We would report either outcome: a crosswalk with quantified uncertainty, or evidence that none exists for this construct.

Linking functions, purification outcomes, and the leave-one-event-out sensitivity are outputs of the pre-specified application, conditional on its Steps 1 to 4.

7 Step 6: The Cost of Assuming Equivalence in Simulation

The final step quantifies the cost of assuming equivalence, an assumption the field currently makes without testing it. Unlike Steps 1 to 5, it does not require the corpus at all, because the ground truth is constructed: we simulate platforms whose measurement properties are known exactly, and ask how badly the standard index construction misreads them. The naive index pools posts across platforms, the construction canonical since O'Connor et al. [2010]; the dissertation itself built such an index (Figure 7), and its decomposition (Figure 8) shows the trend and seasonal components an analyst would report from it.

Design. A latent daily sentiment series carries a known negative shift at an anchor event and recovers gradually. Each platform observes the same series through its own measurement function: an affordance bias shifts every expression, platform-specific thresholds discretise the result into

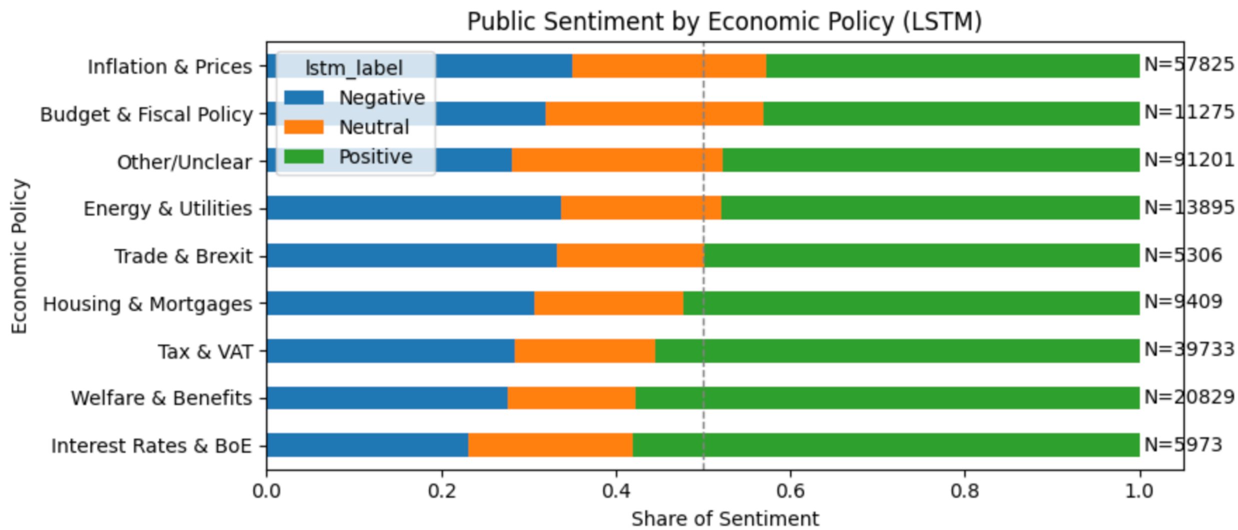

Figure 6: The same corpus rescored with the BiLSTM (dissertation Figure 7.13, p. 42). The broad category ordering is nearly preserved, with adjacent categories swapping rank; measured shares are not preserved. Compare Figure 2.

negative, neutral, and positive, and expression noise is added; platform volume shares also drift over time, so composition changes as it did in the post-API record [Davidson et al., 2023, Goanta et al., 2026]. Two indices are then computed daily. The naive index is the pooled mean of coded values across platforms. The adjusted index estimates each platform's latent value by inverting that platform's measurement function, then pools the estimates weighting each platform by its daily post volume. In the application the measurement parameters come from Step 3; in the simulation they are known, so the exercise isolates what adjustment can recover in principle and what pooling destroys.

Quantities reported. We compare the two indices on what an analyst would actually report: the estimated sentiment shift at the event and the day-by-day sign of aggregate sentiment; the recovery trajectory is shown for a single run in Figure 1. The generating parameters were fixed before the replications were run and ship with the code. The event shift is estimated, for the latent series and for each index alike, as the mean over the ten days after the event minus the mean over the ten days before it; under the generating parameters, a step of -0.5 decaying with a ten day time constant, this contrast is -0.332. Across 500 replications (seeds 20261201 to 20261700), the naive index overstates it by 0.042 on average (2.5th to 97.5th percentiles across replications, 0.032 to 0.051 in absolute size), an error of roughly an eighth of the true shift, while the adjusted index recovers it with negligible bias (mean error 0.001, under half of one per cent of the true shift).

The naive index also reports the wrong sign of aggregate sentiment, positive where the latent truth is negative or the reverse, on 13.8 of 60 simulated days on average (percentiles 11 to 17): for nearly a quarter of days, an analyst reading the pooled index would tell a policymaker that net sentiment points the wrong way. The adjusted index averages under one wrong sign day per run (mean 0.8, percentiles 0 to 2), so the naive count is not an artefact of days on which the latent series sits near zero.

These figures are properties of the simulated configuration rather than bounds, and the two errors have different sources: the shift misstatement arises mainly from taking the discretised

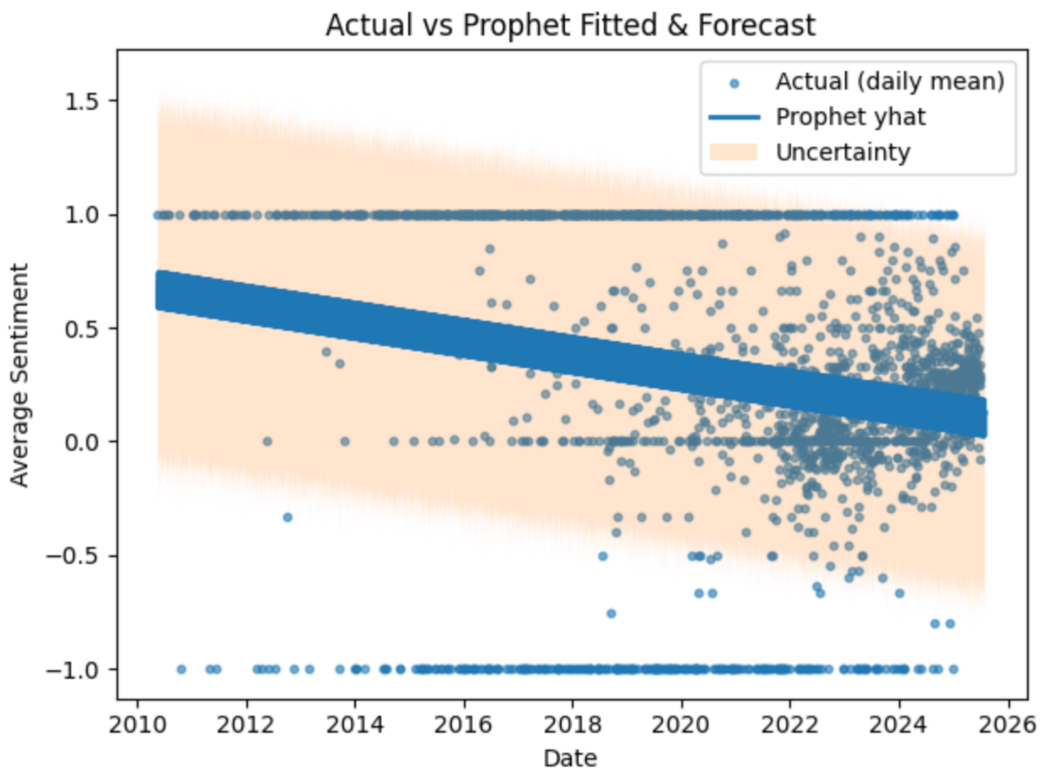

Figure 7: The naive pooled index in practice: dissertation-stage daily mean sentiment pooled across platforms, with the Prophet fit and forecast band (dissertation Figure 7.7, p. 39). Several problems with the construction are visible here: days with a single post pin the daily mean at plus or minus one, no support threshold or volume weighting is applied, the uncertainty band ignores the bounded scale, and the axis spans 2010 to 2026, beyond any stated collection window. The simulation isolates the measurement component of exactly this construction.

three point scale at face value and persists even when all platforms share one unbiased symmetric measurement function, whereas the wrong sign days all but vanish in that case and are specifically the cost of non-equivalence.

What is deferred to the application. Corpus-based variants of this exercise, reweighting platform shares to mimic a platform losing API access, and the forecast-error comparison that feeds both indices to a Prophet model as the dissertation's forecasting stage did, require the application corpus and are pre-specified for the follow-up study. Corrections for label error [Egami et al., 2023] would leave the simulated quantity unchanged in either case: they address disagreement between classifier and criterion, whereas the artefact here arises between platforms.

8 Limitations

Our claims are subject to five limitations. First, the selection confound. Users choose their platforms, and the same-event design controls topic and time, not who is posting. Any non-invariance we detect therefore bundles population composition with measurement mode; because the two can offset as well as compound, the bundle cannot be read as a bound on the pure mode effect, only as

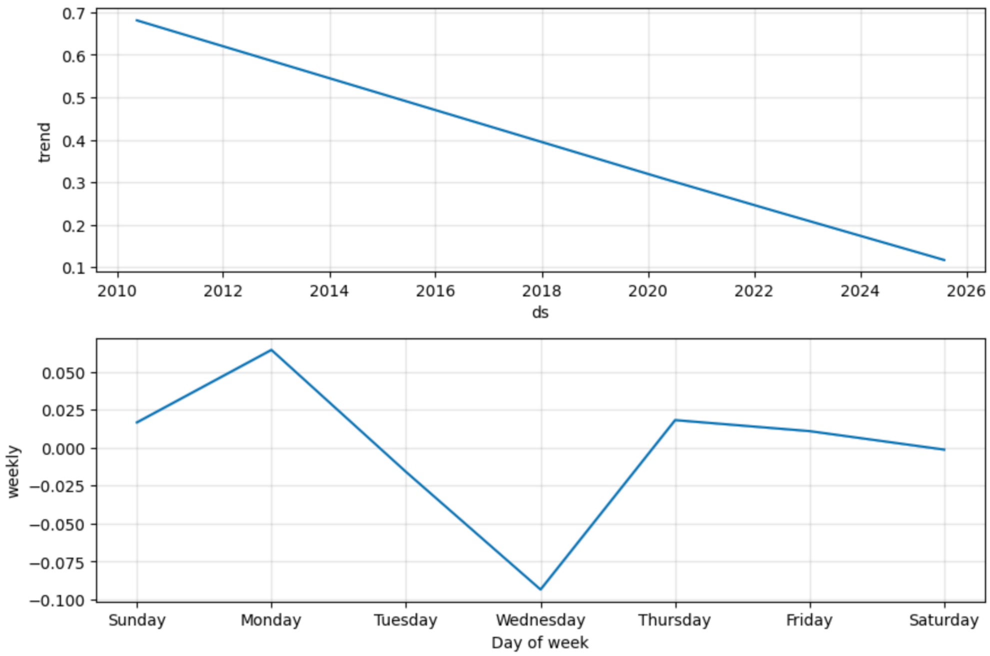

Figure 8: Dissertation-stage Prophet decomposition of the pooled index (dissertation Figure 7.6, p. 39): a declining long-run trend and a midweek dip that the simulation treats as a candidate composition artefact, equally consistent with weekday differences in platform mix and with mood itself.

the platform-level difference in how the construct is measured [Sen et al., 2021]. The convergence criterion [Czarnek and Stillwell, 2022] removes classifier artefacts from this bundle; it cannot remove selection. Separating the two would require within-user cross-posting data, which we do not have.

Second, there is the question of platform scope. We study four platforms, and the anchor-density audit may narrow the analysis to the subset with sufficient same-event coverage. All findings are conditional on the retained platforms and events. Exclusion of a sparse platform is a data limitation, not evidence that it measures equivalently. Likewise, following Robitzsch and Lüdtke [2023], we read invariance results as diagnostic evidence about this corpus, not as a general warrant to pool.

Third, the construct itself. Our instrument measures expressed sentiment, not policy support. Negative sentiment in a thread about inflation need not be opposition to any policy [O'Connor et al., 2010]. Our equivalence results concern the sentiment measure itself; any mapping to attitudes would need separate validation.

Fourth, a single country and language. All data are UK-focused and English. Machine scoring is language-dependent [Nogara et al., 2025], and platform affordance mixes differ across national ecologies, so we make no cross-country claim.

Finally, provenance. The dissertation underlying this paper reported several internally inconsistent statistics: per-platform counts summing to roughly 395,000 against a 279,000 headline (pp. 21-25), a forecast MAE of approximately 0.02 in the text but 0.05 in the goal table (pp. 39, 46-47), post-calibration ECE of 0.05 in the methodology but 0.03 in the results while the reliability diagrams themselves embed ECE values of 0.534 for BERT and 0.012 for BiLSTM, contradicting both (pp. 30, 36-37), manual labelling coverage of 0.07% in one chapter and approximately 0.5% in another (pp. 25, 45), an accuracy figure appearing as both 88.06% and approximately 0.89 (pp. 35, 50), and a forecast figure titled with a platform the study excluded (p. 38). The execution records add two more: the dissertation text reports 85 per cent model agreement where the executed notebook computes 88.1 per cent, and the reproducibility appendix describes a temporal train-test split and older library versions where the executed code applies a stratified random 90/10 split with seed 42 under newer libraries. The dissertation-stage figures reproduced in this draft are included as evidence of the instrument as it stood; the application will produce fresh counterparts of every figure from its own corpus, and the inconsistent reliability diagrams are deliberately not reproduced. Every dissertation-stage statistic reported in this paper is confined to what the execution records, the code, the pipeline logs, and the fully executed notebook, establish, and is labelled as such; quantities those records do not settle, the per-platform composition and the collection window among them, are not estimated here. The empirical application proceeds by fresh collection under the pre-specified design, scoped to the dense anchor windows the design requires, with provenance logged from the first request.

9 Ethics Statement

This is an observational study of public posts on public policy topics. Collection was approved under the University of Portsmouth ethics review process (School of Computing, Faculty Ethics Committee route; certificate TETHIC-2025-111094, application 27 May 2025, supervisor sign-off 1 June 2025), which classified the project as low risk. The study complies with UK GDPR: no private, deleted, or direct-message content was collected; released artefacts contain no personal identifiers; and quoted examples are paraphrased. The dissertation-stage criterion set comprised 200 manually labelled posts balanced across platforms, with inter-annotator agreement of approximately 0.78 (Cohen's kappa; dissertation p. 28). [TODO: One sentence on who annotated the dissertation-stage criterion set.] The follow-up application's collection and annotation will carry their own ethics cover. The main foreseeable misuse is reading our diagnostics as a ranking of platforms by accuracy; they test equivalence and do not score quality, and we caution against that reading.

10 Conclusion

Until shown otherwise, the platform should be treated as part of the measurement instrument. We formalise that claim and specify its test, adapting the psychometric equivalence toolkit, we believe for the first time, to machine-labelled policy sentiment with the platforms as the groups. The simulation shows the stakes are real: across 500 replications of a single configuration fixed in advance, a pooled index that takes platform scores at face value misstates a known event reaction by roughly an eighth and, because the platforms measure through different functions, reports the wrong sign of aggregate sentiment on nearly a quarter of days; a measurement-adjusted index avoids both errors. Whatever the application then finds, something useful follows: if invariance holds we will know where pooling can be defended, and if it fails we will have an estimate of where, and by how much, pooling misleads.

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A Robustness Battery

Every headline result of the pre-specified application is paired with the checks below; the simulation of Section 7 reports replication intervals under its fixed generating parameters; sensitivity to those parameters is deferred to the follow-up study.

Invariance and DIF. Categorical-indicator models with threshold constraints, judged on effect-size criteria rather than chi-square. Alongside the constrained models we run alignment [Muthén and Asparouhov, 2018] and anchor-free DIF [Chen et al., 2023], so no conclusion depends on a hand-picked anchor set.

Classifier-artefact control. The sentiment-indicator test suite is replicated in full with a non-lexicon transformer; the NRC emotion indicators, which derive from a lexicon, are assessed in the lexicon arm alone and any finding on them is labelled lexicon-conditional. A result counts as a platform mode effect only when both classifier families show it [Czarnek and Stillwell, 2022]; a pattern present under one family alone is recorded as a classifier artefact.

Linking sensitivity. Leave-one-event-out re-estimation of linking constants over the pre-identified fiscal-event anchors, with the September 2022 mini-budget as the main anchor; both 48-hour and 2-week event windows; anchor purification fixed and applied before any estimation.

Uncertainty and calibration. Bootstrap confidence intervals on all indices; McNemar tests for classifier comparisons; temperature-scaled calibration checked per platform. At dissertation stage the executed notebook records McNemar $b = 22{,}429$, $c = 5{,}680$, $p < 0.001$, and a bootstrap macro-F1 difference of 0.067 (95 per cent interval 0.065 to 0.068) between the two classifier families, against the VADER-seeded reference labels.

Annotation reliability. Expansion of the human criterion sample from the dissertation's 200 balanced posts (Cohen's kappa approximately 0.78; dissertation p. 28) to a 1,000 to 1,500 comment probability sample with a second-annotator subset; the agreement statistics are outputs of the application.

B Dissertation-Stage Diagnostic Outputs

The dissertation-stage per-class diagnostics are reproduced here in their original form (Figures 9 and 12), recording the state of the instrument at dissertation stage; they are console outputs rather than typeset figures; the application will produce typeset counterparts, with fresh reliability diagrams, from its own corpus. The application's figure standard is fixed now: console reports become typeset tables that name their reference labels; paired figures share category order; the BiLSTM is named consistently; legends sit outside the axes in plain English; unsigned rates use a sequential palette; axes are limited to the data range; interval levels are stated; output is vector. Both classifiers struggled most with the neutral class; the expanded criterion sample is designed to measure this pattern properly. Figure 10 shows the executed notebook's agreement tally between the two families, the source of the 88.1 per cent figure in Section 5, and Figure 11 an example of the per-comment NRC match counts from which the Step 3 binary indicators derive.

```
📊 Classification Report (per class):
              precision    recall  f1-score   support

    negative       0.86      0.86      0.86     85305
     neutral       0.90      0.86      0.88     52383
    positive       0.89      0.91      0.90    117758

    accuracy                           0.88    255446
   macro avg       0.88      0.87      0.88    255446
weighted avg       0.88      0.88      0.88    255446


🔢 Confusion Matrix:
          pred_neg  pred_neu  pred_pos
negative     73486      2187      9632
neutral       3980     44873      3530
positive      8113      3046    106599
```

Figure 9: Dissertation-stage per-class report and confusion matrix, fine-tuned BERT (dissertation Figure 7.1, p. 36). Class supports total 255,446; the reference labels are the VADER-seeded corpus labels, so these diagnostics measure agreement with the seed lexicon, not accuracy against the human criterion.

C Collection Keyword Scheme

The dissertation's auditable seed rules (p. 18), reproduced here in typeset form; the pre-specified application fixes this scheme ex ante for its fresh collection. Six topics, each with a keyword list and, for the first two, regular-expression hints: taxation and fiscal policy (tax, VAT, income tax, NI, national insurance, HMRC, budget, chancellor, Finance Bill, fiscal, levy); inflation, cost of living and prices (inflation, CPI, prices, cost of living, food prices, fuel prices, energy bills, price cap, BoE, interest rate); welfare, benefits and Universal Credit (welfare, benefits, Universal Credit, UC, DWP, benefit cap, means-tested, child benefit); energy and utilities (energy price guarantee, Ofgem, electricity bill, gas bill, standing charge, fuel duty); housing and mortgages (mortgage, rent, landlord, housing benefit, stamp duty, help to buy); and the labour market (minimum wage, national living wage, pay rise, pay freeze, strike, unemployment). When more than one topic matched, priority ran taxation, then inflation, welfare, energy, housing, labour. Negation terms within three tokens of a polarity keyword flagged the post for review. A gazetteer of institutions (HM Treasury, Bank of England, DWP, Ofgem, OBR) and role mappings (Chancellor, Prime Minister, Shadow Chancellor) supplied entity hints.

D Data and Code Availability

Code, configuration, annotation guidelines, and label sets are available at [TODO: repository URL]; the application adds aggregated per-event statistics. The dissertation stage ran on Google Colab (Python 3.11); the executed notebook's environment (torch 2.8, transformers 4.55, spacy 3.8, nltk 3.9, pandas 2.2) differs from the versions listed in the dissertation's reproducibility appendix (p. 19), and the application will pin and report its environment exactly. The executed split was stratified random, 90 per cent train and 10 per cent test (229,901 and 25,545 comments, seed 42), not the temporal split the reproducibility appendix describes; the application will use and report a

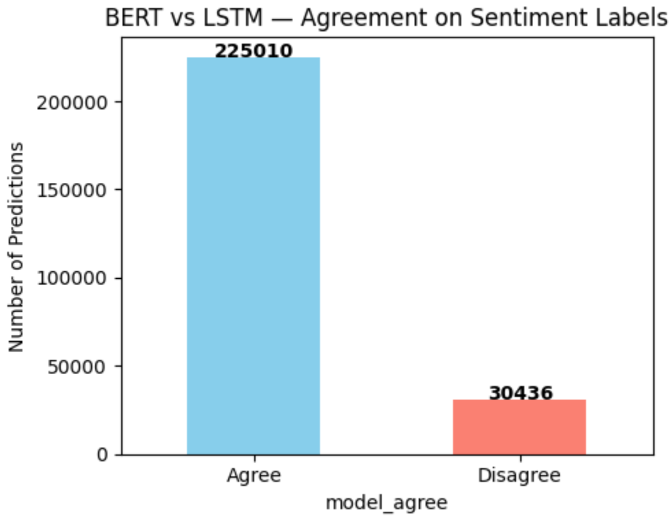

Figure 10: Agreement between the two classifier families over the classified corpus, from the executed notebook: 225,010 of 255,446 comments (88.1 per cent). The dissertation text's figure of 85 per cent is superseded.

temporal split alongside the random one. Raw post text is not redistributed, in line with platform terms; post identifiers are released where terms permit rehydration. Released artefacts contain no personal data. Compute used: a single NVIDIA T4 GPU (16 GB) on Google Colab Pro for the dissertation-stage models (dissertation p. 19); run times were not logged. The simulation runs in minutes on a laptop.

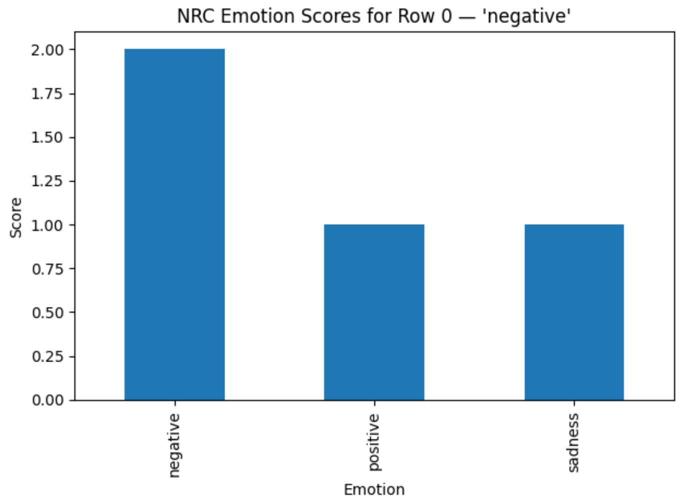

Figure 11: Per-comment NRC lexicon match counts for one example (executed notebook). These raw counts, not the single dominant label, are the basis of the Step 3 binary emotion indicators.

```
📊 Classification report (full data):
              precision    recall  f1-score   support

    negative       0.81      0.74      0.77     85305
     neutral       0.81      0.83      0.82     52383
    positive       0.82      0.87      0.84    117758

    accuracy                           0.82    255446
   macro avg       0.81      0.81      0.81    255446
weighted avg       0.81      0.82      0.81    255446


🔢 Confusion matrix (full data):
          pred_neg  pred_neu  pred_pos
negative     62837      4449     18019
neutral       4435     43270      4678
positive     10069      5587    102102
```

Figure 12: Per-class report and confusion matrix for the BiLSTM at dissertation stage (Figure 7.2, p. 36). The reference labels are again the VADER-seeded corpus labels, not the human criterion.